Final Project for Control System Design and Simulation

Name: ____________________________

Student ID: _______________________

Class: 23 Automation Class 2

Suqian University • College of Mechanical and Electronic Engineering

2026.04.27

Requirements for the Final Project

The Lorenz Attractor

$$
\begin{aligned}
\dot{x}_1 &= -8/3x_1 + x_2x_3 \\
\dot{x}_2 &= -10x_2 + 10x_3 \\
\dot{x}_3 &= -x_1x_2 + 28x_2 - x_3
\end{aligned}
$$

where， $x_1(0) = 0$, $x_2(0) = 0$, $x_3(0) = 1e - 10$。

Simulate the Lorenz system using five methods

- M-file
- Simulink blocks（Simplifying any module is not allowed.）
- Fcn module
- Matlab function module
- S-function

State curves and phase plane curves of the Lorenz system.

Using the above five methods, obtain the state curves and phase plane curves of the Lorenz system under the required initial conditions. The state curves and phase plane curves are required to be smooth, with a simulation time of 100 seconds. Retain all data within the 100-second simulation period.

Grading Criteria

- The main body of the final project report must be completed in Word (following the requirements of the provided Word template). After completing the main body of the report, print it (single-sided printing). At the same time, print this document (including the cover and the project requirements, totaling 3 pages, single-sided printing). Staple the two printed documents together.（This document should come first, followed by the main body of the report.），Also fill in the personal information on the cover page.(Both documents must have the name and student ID filled in.) Submit the documents afterward. Format review accounts for 15 points.
- The five methods required in the main body of the final project report must be described in detail, including the simulation process and various issues encountered during the

simulation (such as unsmooth curves, how to obtain phase curves, etc.), along with the corresponding solutions. The original code, simulation modules, and simulation results (state diagrams and three-dimensional phase curve diagrams) must all be screenshotted. The naming of the diagrams or the original program modules must reflect the individual's surname followed by a numeric identifier. The screenshots of the simulation code and the resulting curves must be clear and legible. This section is worth 40 points.

- All work must be original (completed independently on the computer). Do not arbitrarily copy the results of others. If obvious similarities are found, the involved students will receive zero points for this section. This section constitutes an integrity score of 15 points.
- The obtained state curves and phase plane curves must be smooth, and the data information within the simulation time must be complete. This section accounts for 15 points in simulation quality.
- The final printed version of this project must be submitted by class no later than May 24, 2026 (with all cover page information completely filled out). After collecting all projects, the academic monitors of each class should submit them on time to the Automation Teaching and Research Office, Room A410. This section is worth 15 points.
- The final total score is calculated as 40% based on regular performance (assignments and in-class tests) and 60% based on the final project.

Item Evaluation Content	Report Structure) （15%）	Report Quality （40%）	Originality （15%）	Simulation Quality （15%）	Project Efficiency （15%）
Scores for each evaluation item					

Total： ________________